Chromosoma (2006) 115:437–447
DOI 10.1007/s00412-006-0070-8

RESEARCH ARTICLE

Sequence homogenization and chromosomal localization of VicTR-B satellites differ between closely related *Vicia* species

Jiří Macas · Alice Navrátilová · Andrea Koblížková

Received: 8 March 2006 / Revised: 4 May 2006 / Accepted: 5 May 2006 / Published online: 20 June 2006
© Springer-Verlag 2006

Abstract Satellite sequences of the VicTR-B family are specific for the genus *Vicia* (Leguminosae), but their abundance varies among the species, being the highest in *Vicia sativa* and *Vicia grandiflora*. In this study, we have sequenced multiple randomly cloned VicTR-B fragments from these two species and analyzed their sequence variability, periodicity, and chromosomal localization. We have found that *V. sativa* VicTR-B sequences are homogeneous with respect to their nucleotide sequences and periodicity (monomers of 38 bp), whereas *V. grandiflora* repeats are considerably more variable, occurring in at least four distinct sequence subfamilies. Although the periodicity of 38 bp was conserved in most of the *V. grandiflora* sequences, one of the subfamilies was composed of higher-order repeats of 186 bp, which originated from a pentamer of the basic repeated unit. Individual VicTR-B subfamilies were preferentially located in either intercalary or subtelomeric regions of chromosomes. Interestingly, two *V. grandiflora* subfamilies with the highest similarity to *V. sativa* VicTR-B sequences were located in intercalary heterochromatic bands, showing similar chromosomal distribution as the majority of VicTR-B repeats in *V. sativa*. The other two *V. grandiflora* subfamilies showing a considerable divergence from *V. sativa* sequences were found to be accumulated at subtelomeric regions of *V. grandiflora* chromosomes.

Communicated by I. Schubert

Electronic Supplementary Material Supplementary material is available for this article at http://dx.doi.org/10.1007/s00412-006-0070-8 and is accessible for authorized users.

J. Macas (✉) · A. Navrátilová · A. Koblížková
Institute of Plant Molecular Biology,
Branišovská 31,
České Budějovice 37005, Czech Republic
e-mail: macas@umbr.cas.cz

Introduction

Satellite DNA (satDNA) represents one of the major classes of repeated sequences present in complex eukaryotic genomes. It is composed of basic repeated units (monomers) which are tens to thousands of nucleotides long and occur in the genome in a tandem (head-to-tail) organization. In higher plants, over 160 families differing in their nucleotide sequences have been described so far. Although monomer length varies greatly among the families, the lengths around 165 bp and its multiples are clearly most frequent (Macas et al. 2002). The monomers belonging to the same repeat family are usually very similar in size and sequence and form continuous arrays spanning up to 100 Mbp (Charlesworth et al. 1994; Schmidt and Heslop-Harrison 1998). This represents a major obstacle in their investigation because these regions are extremely difficult to analyze using conventional approaches. Thus, sequence information available for most plant satellites is limited to a few monomers (Macas et al. 2002), and only a little is known about their sequence variability and arrangement of sequence variants on chromosomes and within individual repeat clusters. However, this information is crucial for our understanding of the molecular mechanisms of satDNA amplification and homogenization, which belong to the most dynamic mechanisms in the genome, often leading to the occurrence of highly amplified genus- or even species-specific satDNA families (King et al. 1995; Matyášek et al. 1997; Macas et al. 2000; Dechyeva et al. 2003; Tek et al. 2005).

VicTR-B is a family of satellite repeats specific to the genus *Vicia*. The repeats are AT-rich and their consensus monomer sequence is 38 bp in size, as determined from several sequenced clones and Southern blot analysis. VicTR-B is differentially amplified in *Vicia* species,

ranging from tens up to millions of copies per haploid genome (1C). The degree of VicTR-B amplification correlates with taxonomic classification within the genus, being the highest in *Vicia sativa* and *Vicia grandiflora*, both belonging to the section *Vicia* (Macas et al. 2000). VicTR-B repeats occur in long arrays forming heterochromatic bands on mitotic chromosomes. However, their distribution on chromosomes is not conserved between species (Navrátilová et al. 2003).

In this study, we aimed at detailed comparative analysis of the sequence variability of VicTR-B repeats from two closely related species, *V. sativa* and *V. grandiflora*. Both of these species contain equally amplified VicTR-B repeats reaching millions of copies per 1C. In *V. sativa* (2n=12, 1C=2.3 pg), VicTR-B is mainly located in intercalary bands present on five out of the six chromosome pairs, and only minor clusters are detectable in terminal chromosome regions. On the other hand, *V. grandiflora* (2n=14, 1C=3.4 pg) has a larger portion of the repeats distributed in subtelomeric regions, although a significant part of VicTR-B still occurs as bands within chromosome arms. To obtain representative sets of VicTR-B repeats for analysis, we avoided using PCR for their cloning. Instead, shotgun libraries carrying inserts of 500–800 bp were constructed from sonicated and size-fractioned genomic DNAs and used for isolation of VicTR-B fragments. The insert size selected for cloning allowed reliable sequence determination and assembly, while preserving information about the short-range variation of monomer sequences within cloned repeat arrays. These sequences were then subjected to computer analysis to assess their periodicity patterns and mutual similarities. In addition, long-range distribution of various sequence subfamilies was then investigated using primed in situ DNA labeling (PRINS) on mitotic chromosomes.

Materials and methods

Plant material and cloning procedures

Seeds of *Vicia* species were obtained from the germplasm collection of the Institute of Plant Genetics and Crop Plant Research, Gatersleben, Germany (*V. grandiflora* Scop., accession number VIC741/89) and from the plant breeding station at Horní Moštěnice, Czech Republic (*V. sativa* L. cv. Ebena). Total genomic DNA was isolated from young leaves as described by Dellaporta et al. (1983).

The shotgun genomic libraries were prepared from genomic DNA subjected to sonication, mung bean nuclease treatment, and size fractionation on an agarose gel. The fragments of about 500–800 bp were purified from the gel, treated with polynucleotide kinase in the presence of ATP, and cloned into *Sma*I-cut and dephosphorylated plasmid vector (pBluescript II SK+). Resulting colonies were picked into 384-microwell plates and subsequently gridded onto hybridization membranes using GeneTAC G3 robot (Genomic Solutions). The libraries were screened by colony hybridization with previously isolated VicTR-B clones S1 or GS38 (Macas et al. 2000) as a probe, using the AlkPhos Direct hybridization and detection kit (Amersham). Moderate hybridization and washing stringency (both performed at 45°C) was used to allow for identification of VicTR-B repeats with possibly lower similarity to the probe. Sequencing was performed by the dideoxy chain termination method (Sanger et al. 1977) and the resulting sequences were assembled in the Staden Sequence Analysis Package (Staden 1996). Each clone was sequenced in both directions and only the high-quality readings spanning the entire insert and surrounding polylinker sequences were used. These precautions ensured the correct assembly of sequences, avoiding misalignment of the readings due to the strong similarity of VicTR-B monomers.

Sequence analysis

Unless specified otherwise, VicTR-B sequences were analyzed using scripts implemented for this purpose in Perl and running under the Linux operating system. Sequence periodicity analysis was based on the concept of nucleotide autocorrelation functions, and expressed for a distance of k base pairs and nucleotide X as a difference $C_{XX}(k)=p_{XX}(k)-p_X.p_X$, where p_{XX} is the observed frequency of identical nucleotides X and p_X is the proportion of nucleotide X in the sequence. Thus, a positive value of C_{XX} implies that there are more X–X pairs at distance k than expected by chance (Herzel et al. 1999). The calculations were performed either separately for individual clones or averaged for a set of clones. Multiple sequence alignments were calculated using ClustalW version 1.83 (Thompson et al. 1994) and displayed using Boxshade 3.3 (created by Kay Hofmann and Michael D. Baron, available from ftp://www.isrec.isb-sib.ch/pub/software/unix/boxshade/). Pairwise similarities of monomers were extracted from ClustalW output and transformed into a data set, which served as input to Mgraph (created by Louis Gonzalez and Christine Deroo, available from http://www-loa.univ-lille1.fr/Mgraph/) to generate a dot-plot graph.

Primed in situ DNA labeling (PRINS)

Prior to performing PRINS on chromosomes, the specificity of oligonucleotide primers designed for discrimination of different VicTR-B subfamilies was tested using PCR reactions with individual VicTR-B clones as templates. In these reactions, a single VicTR-B primer was always used

Table 1 Sequence subfamilies of VicTR-B repeats

Species	Subfamily	Definition	Number of clones	Total monomers[a]	Average monomer similarity (%)		Accession numbers	Clones including heterogenous sequences[b]
					Within clones	All monomers		
V. sativa	–	Conserved monomer length (38 bp), no distinct subfamilies	19	279	86.2	85.9	DQ139346, DQ139347, DQ139349–DQ139365	c616, c626, c629
V. grandiflora	Ca	Conserved monomer length (38 bp), similar to *V. sativa* sequences	13	152	83.8	82.1	DQ139366–DQ139378	c606 (retroelement-like sequence), c770, c786, c843, c844, c845
	Cg	Conserved monomer length, sequences differ from Ca subfamily	8	114	87.0	86.8	DQ139379–DQ139386	c562 (telomeric repeat)
	V	Variable monomer length, occurrence of 186-bp HORs	10	160	77.9	78.3 (89.1)[c]	DQ139390–DQ139399	
	D	Conserved monomer length but contains large deletions	3	46	87.5	87.4	DQ139387–DQ139389	

[a]Number of complete monomer sequences extracted based on similarity to previously published 38-bp consensus (Macas et al. 2000)
[b]See electronic supplementary material Tables 1 and 2 for more information about clones containing other sequences in addition to VicTR-B
[c]This value gives average similarity among 186-bp HORs

in combination with a vector-derived primer (either T3 or T7 primer, depending on the orientation of cloned VicTR-B sequences within the vector). PCR was performed in 30 µl volume of 1× PCR buffer, 2 mM $MgCl_2$, 0.2 mM deoxyribonucleotide triphosphates, 0.2 µM primers, 1.2 U of *Taq* polymerase (Promega), and 0.2 ng of template plasmid DNA. The reaction involved 25 cycles of 94°C for 1 min, 57–65°C for 1 min, and 72°C for 1 min, preceded by an initial denaturation (94°C, 2 min) and followed by a final extension step (72°C, 10 min). Amplification products were analyzed using electrophoresis on ethidium bromide-stained 2% agarose gels.

PRINS reactions were performed on squash preparations of mitotic chromosomes prepared from synchronized root tip meristems as described by Navrátilová et al. (2003). The reaction mix (65 µl) contained 1× PCR buffer, 2 mM $MgCl_2$, 0.1 mM each of 2′-deoxy-ATP, 2′-deoxycytidine 5′-triphosphate, 2′-deoxy-GTP, 0.017 mM 2′-deoxythymidine 5′-triphosphate, 0.008 mM Alexa-fluor-488–2′-deoxyuridine 5′-triphosphate (Molecular Probes), 0.2 µM primer, and 5.2 U of *Taq* polymerase (Promega). Only a single primer was used in each reaction, and the temperature profile consisted of DNA denaturation at 94°C for 3 min, primer annealing at 57–65°C for 5 min, and primer extension at 72°C for 30 min (the heating rate between the annealing and extension steps was 0.1°C/s). Primer sequences and annealing temperatures are listed in Table 2. Following PRINS reaction, the preparations were washed in 4× saline-sodium citrate, counterstained with 4,6-diamino-2-phenylindole, and observed using a Nikon Eclipse-600 epifluorescence microscope equipped with a CCD camera. The signals were collected using appropriate filter sets and LUCIA software (Laboratory Imaging).

Results

Following screening the shotgun libraries with VicTR-B probe, 20 and 34 randomly picked positive clones were sequenced for *V. sativa* and *V. grandiflora*, respectively. The majority of these clones were entirely composed of VicTR-B repeats, although three *V. sativa* and seven *V. grandiflora* clones also included other sequences in addition to VicTR-B. Except for one *V. sativa* clone (c615), which contained a highly degenerated sequence and was excluded from further analysis, all VicTR-B repeats were arranged as regular arrays composed

Table 2 Primers used for PRINS reactions, their specificity for individual VicTR-B subfamilies, and results of PRINS labeling of *V. grandiflora* mitotic chromosomes

Primer	Sequence (5′–>3′)	Annealing[a] (°C)	Specificity[b]				PRINS labeling (positions of signals on chromosomes)	Figure
			Ca	Cg	V	D		
VG-2R	TTCATCWGATTTTCTGAAGACTATTSAC	62	(+)	−	++	+	Subtelomeric	5b
VG-6F	TCCGATGAAATTTGAAGTSAATAGTC	62	−	−	+	−	Subtelomeric	5c
VG-5F	GTGAACGTAATTCTTAACACARTCACA	62	−	−	+	−	Subtelomeric	
VG-5FK	GTGAATATAAGTCTTSAGAAAATCTGG	62	++	(+)	(+)	++	Intercalary bands (strong signals), weak subtelomeric signals on some chromosomes	5d
VG-3R	TTCACCATATTTTCTCAWGATTTATGCC	65	−	++	−	−	Subtelomeric	
VG-7F	TATGGTGAAATTTGAAGTGGGC	62	(+)	++	−	−	Subtelomeric	5e
VG-7FK	TCTGGTGAAATTTGAAGTGAAT	60	+	−	++	+	Both, subtelomeric and intercalary signals	5f
VG-4F	GAATATAAGTCTTCRGAAAATCGAAG	62	−	−	−	+	Intercalary dots on two chromosomes pairs, very weak signals	

[a] Annealing temperature used in PRINS reactions

[b] Ability of the primers to amplify sequences from individual VicTR-B subfamilies, as assessed by PCR with cloned VicTR-B repeats [++ very good, + good, (+) no or inefficient amplification in most clones, − no amplification]. Detailed results of the reactions with all available clones are given in electronic supplementary material Table 3

of units with a strong similarity to the previously described 38-bp monomer (Macas et al. 2000). A total of 279 such monomer sequences could be identified in *V. sativa* and 472 in *V. grandiflora* clones (see electronic supplementary material Tables 1 and 2 for detailed descriptions of the clones).

Periodicity analysis reveals conservation of VicTR-B monomer lengths in *V. sativa* and higher-order repeats in *V. grandiflora*

Nucleotide autocorrelation functions (Herzel et al. 1999) measuring the excess of pairs of identical nucleotides (A–A, T–T, G–G, and C–C) at various distances were used to visualize periodicities within VicTR-B sequences. In *V. sativa*, this analysis confirmed the predominant sequence periodicity of 38 bp and its multimers, appearing as prominent peaks in all four nucleotides (Fig. 1a–d). Detailed analysis of individual clones revealed that four of them displayed alternating smaller and higher positive peaks (Fig. 1e), suggesting an emergence of a more complex pattern caused by stronger homogenization of dimer sequences. On the other hand, the remainder of the clones displayed a pattern of gradually decreasing positive peaks, indicative of no or weaker homogenization of higher-order structures (Fig. 1f).

Periodicity analysis and visual inspection of *V. grandiflora* clones revealed their much higher diversity than that of *V. sativa*. Three groups of clones could be recognized based on the periodicity of their VicTR-B sequences: (1) 21 clones included repeat arrays with conserved monomer lengths of 38 bp, similar to those found in *V. sativa* (Fig. 1g). (2) Three other clones also contained regular 38-bp arrays, but these were occasionally disrupted by large (11 bp) deletions. The deletions always included the same sequence region and occurred once per 5, 7 or 9 monomers. (3) Ten clones were composed of monomers of varying lengths (mostly 35, 37, or 38 bp) alternating in regular patterns. As is evident from their computer analysis (Fig. 1h), the major periodicity peak of these sequences was at 186 bp. This higher-order repeat (HOR) corresponded to a pentamer of the basic 38-bp unit, shortened by a deletion of 4 bp. It should be noted that VicTR-B arrays within individual clones were always uniform with respect to their periodicity, so that an entire array could always be assigned to either of the three groups defined above. The results of periodicity analysis corresponded to restriction patterns of VicTR-B repeats detected on Southern blots of *V. sativa* and *V. grandiflora* genomic DNAs digested with various restrictases (Fig. 2). The patterns of *V. sativa* VicTR-B repeats indicated absence of restriction sites or showed ladders of bands with spacing corresponding to monomer size of 38 bp. On the other hand, the presence of HORs in *V. grandiflora* was evident in most enzymes tested as ladders of prominent bands differing by 186 bp.

Increased sequence variability and occurrence of VicTR-B subfamilies in *V. grandiflora*

Pairwise and multiple sequence alignments were used to investigate similarities between VicTR-B monomers within and among individual clones. A population of *V. sativa* VicTR-B sequences was found to be relatively uniform,

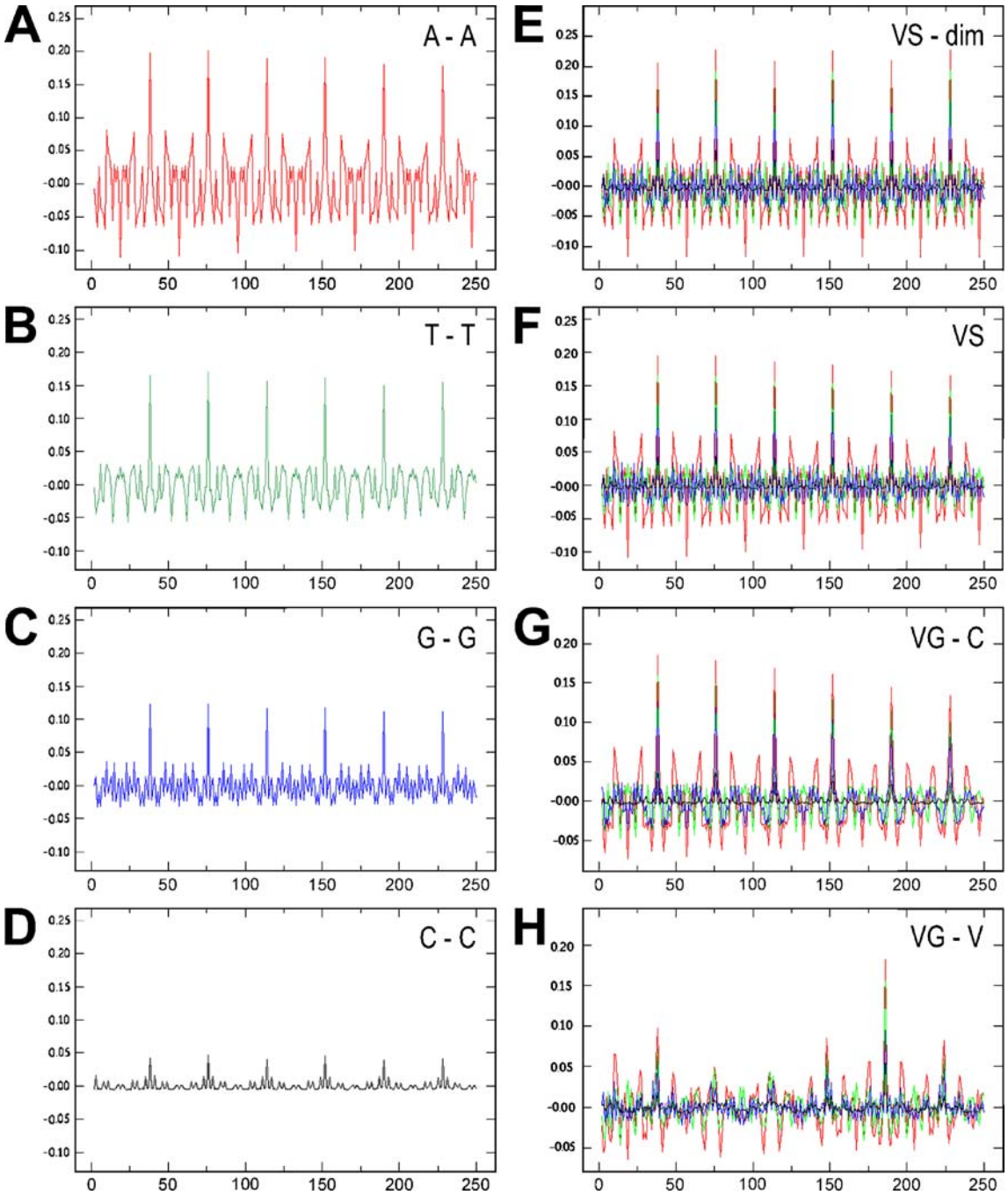


Fig. 1 Periodicity analysis of VicTR-B sequences using nucleotide autocorrelation functions. This analysis measures the excess of pairs of identical nucleotides at various distances (2–250 bp), displaying sequence periodicity as regularly spaced peaks. The results are shown either separately for individual nucleotides (**a**–**d**) or combined for all four nucleotides (**e**–**h**, the nucleotides are distinguished by the same colors as in **a**–**d**). The analysis was performed either for all *V. sativa* sequences (**a**–**d**), or separately for the clones c613, c620, c621, and c622 (**e**) and the rest of *V. sativa* sequences (**f**). Sequence periodicity of *V. grandiflora* is shown on graphs **g** (subfamilies Ca and Cg) and **h** (subfamily V)

with average sequence similarity between all monomers very close to the value obtained for monomers within individual clones (85.9 and 86.2%, respectively). A graphical representation of mutual similarities of all monomers (Fig. 3) shows that this reflects a high degree of sequence homogenization and a lack of distinct subfamilies within *V. sativa* sequences. The distribution of mutations within monomer sequences was not random but it mostly involved variable nucleotides at the 3′ end and also several single positions within the monomers (Fig. 4). In some

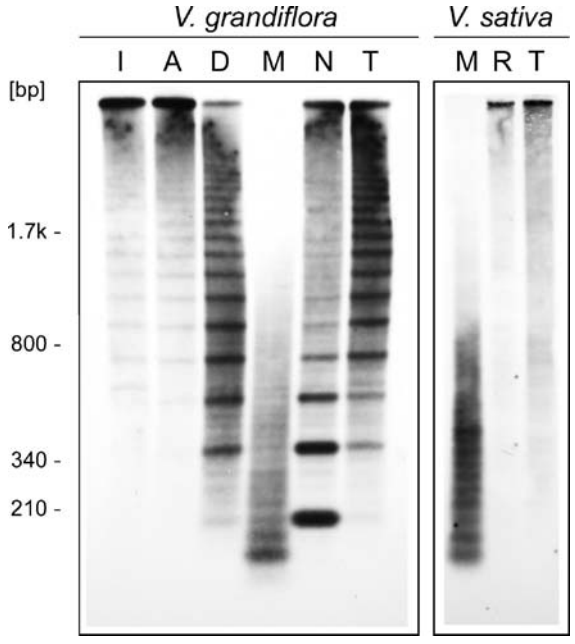


Fig. 2 Restriction patterns of VicTR-B repeats in *V. grandiflora* and *V. sativa*. Southern blots of genomic DNAs digested with *Eco*RI (*I*), *Alu*I (*A*), *Dra*I (*D*), *Mse*I (*M*), *Nla*III (*N*), *Rsa*I (*R*), or *Taq*I (*T*) were hybridized with VicTR-B probe. In *V. grandiflora*, ladders of prominent bands differing by 186 bp were detected in most enzymes tested. The only exception was *Mse*I, producing ladder with band spacing corresponding to the 38-bp monomers. The same pattern was detected in *Mse*I-digested DNA of *V. sativa*; however, other enzymes did not produce any distinct restriction patterns, as shown for *Rsa*I and *Taq*I

clones, the same mutations occurred with a period of two or more monomers; this was the most evident for clones which displayed strong dimer periodicity, as shown above (Fig. 1e; see also electronic supplementary material Fig. 1 for multiple alignment of all *V. sativa* sequences).

Analysis of *V. grandiflora* VicTR-B sequences revealed much higher variability than in the case of *V. sativa*. This was in part due to the variability of monomer lengths, especially in clones containing the 186-bp HORs. However, even within the group of clones with the conserved 38-bp monomer lengths, there were two subfamilies differing considerably in their nucleotide sequences. Thus, based on their periodicity and sequence divergence, four basic subfamilies of VicTR-B repeats were distinguished (Table 1 and Fig. 4). Average similarities between monomers within individual clones compared to those within the subfamilies were, in all cases, very similar, suggesting that the subfamilies were relatively homogeneous. As expected, subfamily V displayed low similarity of its 35–38 bp monomers (78.3%); however, 186-bp HORs gave a much higher score (89.1%), proving that these higher-order units are a preferred target for sequence homogenization (Figs. 4e and electronic supplementary material Figs. 4 and 5). As is evident from Fig. 3, individual subfamilies displayed various degrees of sequence divergence. For example, subfamilies Ca and D were similar to each other (average similarity of 84.2%) as well as to *V. sativa* sequences (82.4 and 83.8% for Ca and D, respectively). On the other hand, subfamily Cg was the most diverged one, and its average similarity to sequences from other subfamilies did not exceed 73%.

Individual VicTR-B subfamilies are preferentially located in either subtelomeric or intercalary bands on *V. grandiflora* chromosomes

Distribution of VicTR-B repeats on *V. grandiflora* chromosomes was studied using PRINS with oligonucleotide primers designed to discriminate individual sequence subfamilies (Table 2). This was achieved by targeting 3′ ends of the primers into specific sequence variants or into positions including deletions. Prior to performing PRINS, the primer specificity was tested in PCR reactions using cloned VicTR-B sequences as a template. As shown in Fig. 5 and summarized in electronic supplementary material Table 3, using optimized reaction conditions allowed selective amplification of defined subsets of VicTR-B repeats, depending on the primers used. Employing these

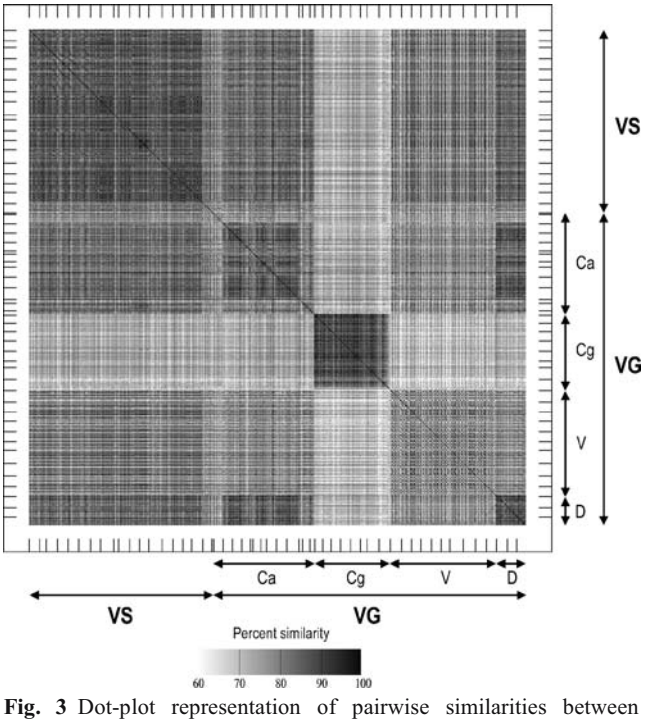


Fig. 3 Dot-plot representation of pairwise similarities between VicTR-B monomers. A total of 751 monomers are ordered according to their position within the clones (individual clones are separated by ticks) and grouped based on the species and subfamilies as indicated (VS *V. sativa*, VG *V. grandiflora*). Percent similarity scores of all possible pairwise alignments are shown as *dots of different shades of gray*, corresponding to the scale given below the dot-plot



Fig. 4 Sequence logos showing conservation of nucleotides in VicTR-B repeats from *V. sativa* (**a**) and *V. grandiflora* subfamilies Ca (**b**), Cg (**c**), D (d), and V (**e**). A complete 186-bp HOR is shown for subfamily V and its five subrepeats are aligned with 38-bp monomers of other subfamilies. The logos were created using the method of Schneider and Stephens (1990) as implemented on the WebLogo server (Crooks et al. 2004), using the multiple sequence alignments (electronic supplementary material Figs. 1, 2, 3, 4) as an input. The nucleotide differences characteristic for individual subfamilies are indicated by the *black background*, deletions in subfamily V sequence are indicated by *dashes*

conditions in PRINS reactions thus resulted in specific labeling and visualization of corresponding sequences on *V. grandiflora* mitotic chromosomes. These signals were compared to labeling obtained using universal primers (Macas et al. 2000) that marked all VicTR-B repeats, showing their location in intercalary and subtelomeric regions of all chromosomes (Fig. 6a). In contrast to this, the primers specific for subfamily V labeled only terminal regions of the chromosomes (Fig. 6b,c and Table 2). Identical results were obtained using three different primers, two of which were targeted in forward and reverse orientation to the subfamily V-specific 3-bp deletion (primers VG-2R and VG-6F), and the remaining one (VG-5F) to the mutations in the region 55–75 of the HOR

sequence (Fig. 4e). The same localization pattern limited to subtelomeric regions was found for subfamily Cg (Fig. 6e), using the primers VG-3R and VG-7F targeted to the specific sequence motif GGC (Fig. 4c, positions 15–17). On the other hand, sequences of the Ca subfamily appeared to be mostly located in intercalary chromosomal bands (Fig. 6d). The primer VG-4F targeted to the region with a 11-bp deletion characteristic for subfamily D produced very weak intercalary signals on two chromosome pairs that were distinguishable on only a part of the scored metaphase spreads. This was most likely due to significantly lower abundance of the subfamily D sequences as compared to other subfamilies.

Discussion

In this study we have characterized the sequence variability of VicTR-B satellites in *V. sativa* and *V. grandiflora*. Although these species are closely related and both contain highly amplified VicTR-B repeats, they differ considerably in the degree of VicTR-B sequence homogenization. It is supposed that satellite repeat evolution is driven by various processes of DNA turnover, including unequal crossover, gene conversion, and amplification (Dover 1982, 2002; Ugarkovic and Plohl 2002). Although the exact molecular mechanisms of these processes, their eventual interactions, and their regulation are still poorly understood, it is well established that they are very dynamic, causing rapid changes in abundance and sequence composition of satDNA. This is evident from the observed rapid emergence or elimination of certain satellite repeats (Langdon et al. 2000; Lim et al. 2004; Skalická et al. 2005; Tek et al. 2005), resulting in profound differences in the abundance of individual satDNA families found between related organisms (King et al. 1995; Matyášek et al. 1997; Macas et al. 2000; Dechyeva et al. 2003). In addition to changes in copy numbers, these mechanisms cause sequence homogenization of satellite repeats within the genome, the process known as concerted evolution. Considering a family of satellite repeats present in a set of species, the consequence of concerted evolution is that sequence similarity of the repeated units is greater within than among species (Elder and Turner 1995). The high degree of sequence similarity of VicTR-B repeats found in *V. sativa* appears to be a result of such homogenization. On the other hand, *V. grandiflora* sequences are much more diverse and can be grouped into several distinct subfamilies. Two of these subfamilies are similar to each other and to *V. sativa* sequences (subfamilies Ca and D), whereas the other two are more diverged (Fig. 3). Thus, it appears that the sequence homogenization process is less efficient in *V. grandiflora*, or that the VicTR-B population in this species undergoes a transition to a new equilibrium.

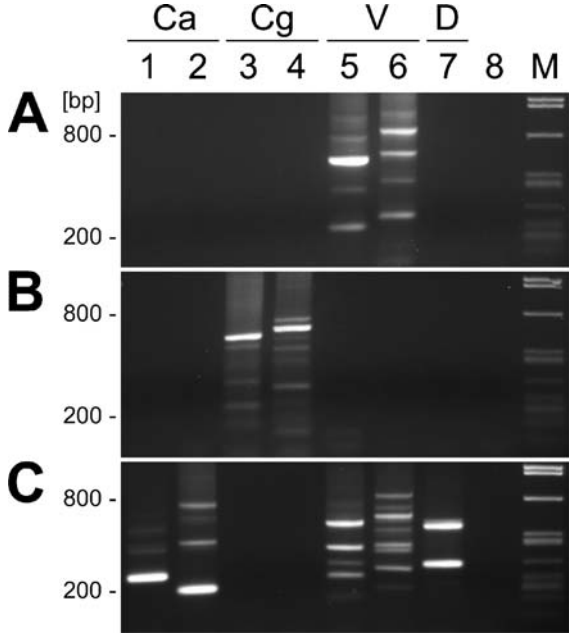


Fig. 5 Specificity of PRINS primers tested in PCR with cloned VicTR-B repeats. The reactions were performed using vector-derived primer T7 and either of the primers VG-6F (**a**), VG-7F (**b**), or VG-7FK (**c**). Clones carrying *V. grandiflora* VicTR-B sequences belonging to different subfamilies (as indicated above the *lane numbers*) were used as templates: c839 (*lane 1*), c840 (*lane 2*), c605 (*lane 3*), c788 (*lane 4*), c609 (*lane 5*), c785 (*lane 6*), and c608 (*lane 7*). *Lane 8* contains negative control (no template) and *lane M* includes DNA fragment length marker (lambda DNA digested with *Pst*I)

It has been reported that monomers from individual satDNA subfamilies are clustered in plant genomes. For example, sequence variants of soybean STR120 satellite were localized to different chromosomes by fluorescence in situ hybridization (FISH) (Morgante et al. 1997). Similar chromosome-specific distribution patterns were found for FISH or PRINS localization of the centromeric satellite AtCon of *Arabidopsis thaliana* (Heslop-Harrison et al. 1999) and the intergenic spacer-like satellite of *V. sativa* (Macas et al. 2003). A recent study analyzing large numbers of centromeric satellite sequences from Arabidopsis and its relatives also revealed their preferential sequence homogenization within the repeat clusters (Hall et al. 2005), which is in agreement with data reported for human centromeric satellites (Schindelhauer and Schwarz 2002). These findings led to the conclusion that spreading of satDNA sequence variants proceeds via fast, intrachromosomal conversion mechanisms and that interchromosomal exchange is a rate-limiting step for global sequence homogenization (Schindelhauer and Schwarz 2002; Hall et al. 2005). In this respect, it is interesting that the distribution of VicTR-B subfamilies in *V. grandiflora* is not chromosome-specific; instead, it correlates with the position of VicTR-B loci within the chromosomes. In addition, a considerable portion of sequence variants

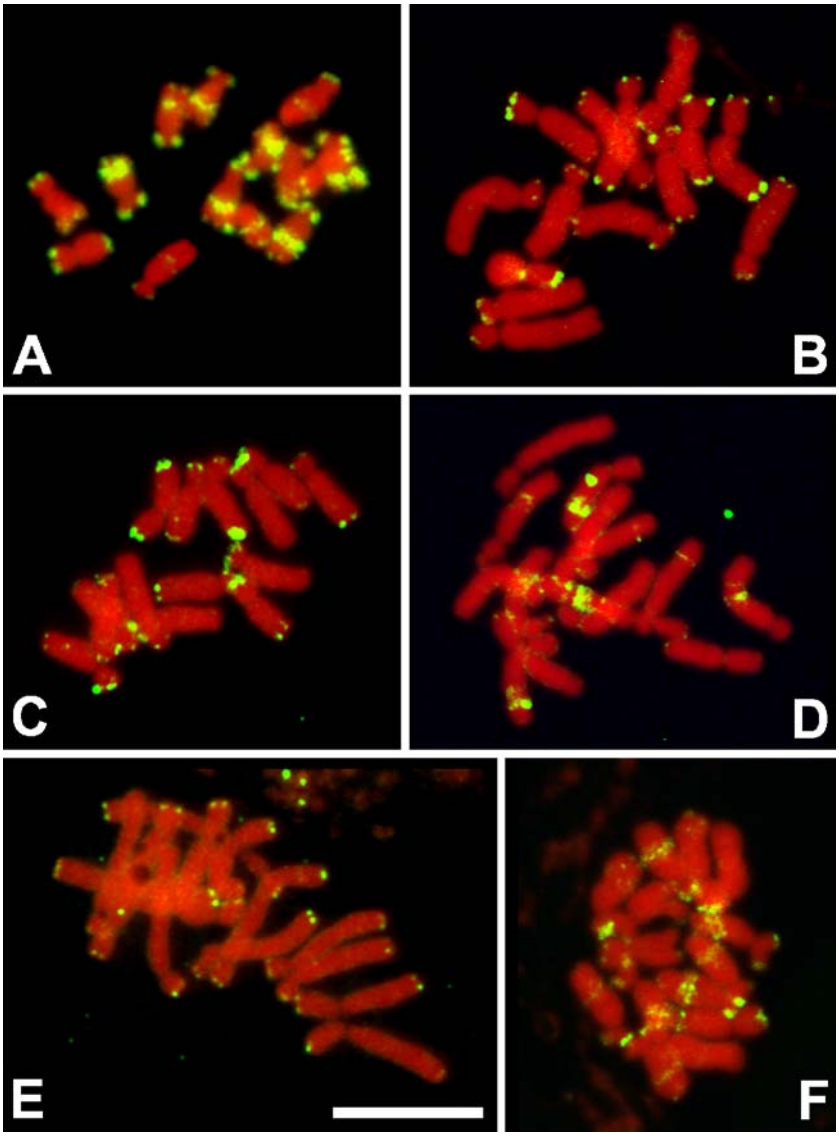


Fig. 6 Localization of VicTR-B subfamilies on *V. grandiflora* chromosomes using PRINS. Labeling of all VicTR-B sequences using universal primers (**a**) is shown for comparison to the results obtained with the primers VG-2R (**b**), VG-6F (**c**), VG-5FK (**d**), VG-7F (**e**), and VG-7FK (**f**). The PRINS signals are *yellow-green*; chromosomes were counterstained with 4,6-diamino-2-phenylindole, which is displayed in *red*. *Bar*=10 μm

represented by subfamilies Cg and V is located in terminal chromosomal regions, whereas intercalary clusters include sequences similar to *V. sativa* repeats. As the most *V. sativa* VicTR-B loci occur within chromosome arms, these findings raise the question of whether increased sequence variability of *V. grandiflora* VicTR-B repeats is a consequence of their higher accumulation in subtelomeric regions. In humans, these regions were found to be hotspots for interchromosomal recombination (Linardopoulou et al. 2005), and also, in plant chromosomes they were found to have a complex long-range structure, probably resulting from extensive rearrangements (Alkhimova et al. 2004). On the other hand, increased recombination was also observed in satellite repeat clusters located within intercalary regions of *Vicia faba* and *Ornithogalum longibracteatum* chromosomes (Schubert et al. 1994; Baranczewski et al. 1997; Pedrosa et al. 2001). Obviously, further studies including a broader range of repeats and plant species are needed to evaluate what the key factors and mechanisms causing sequence diversification of satellite repeats are. In addition to investigating sequence variability with respect to recombination rates and chromosomal localization, other techniques like chromosomal orientation and direction FISH (Meyne and Goodwin 1995) can be used to provide further information about long-range genomic organization of the repeat clusters. This technique has already been successfully applied to *V. sativa* and revealed that most VicTR-B clusters are homogeneous with respect to the orientation of their monomers (Navrátilová et al. 2005).

Periodicity analysis of VicTR-B repeats revealed that most of them are homogenized as 38-bp monomer sequences. However, some of the *V. sativa* sequences showed enhanced homogenization of dimer repeats, and the *V. grandiflora* V family clearly evolved a HOR structure with units originating from a pentamer sequence. The occurrence of HORs or monomers composed of tandemly repeated subunits has been reported for a number of satellites (Ugarkovic and Plohl 2002), including plant repeats TIII15 in *V. faba* (Nouzová et al. 1999), COM2 in Aveneae species (Grebenstein et al. 1996), or RAE730 in *Rumex acetosa* (Shibata et al. 2000). Interestingly, the repeat length of subfamily V HORs (186 bp) falls within the range of 135–195 bp, which is most frequent in plant satDNAs (Macas et al. 2002). It has been pointed out by several authors that this size corresponds to the length of DNA wrapped around a nucleosome particle (Kubis et al. 1998; Schmidt and Heslop-Harrison 1998). Nucleosome phasing has been reported for some plant satellites (Gazdová et al. 1995; Matyášek et al. 1997; Vershinin and Heslop-Harrison 1998), suggesting that nucleosome positioning at certain sequences could provide a constraint for their recombination, which is one of the key processes in satellite repeat evolution (Stephan 1989; Stephan and Cho 1994). However, the occurrence of nucleosome phasing and its exact role in satDNA evolution requires further investigation in a wider range of repeat families, including VicTR-B repeats of *Vicia*.

Acknowledgements We thank Ms. J. Látalová and Ms. H. Štěpančíková for excellent technical assistance. This work was supported by grants GA204/04/1207 from the Czech Science Foundation and AVOZ50510513 from the Academy of Sciences of the Czech Republic. The experiments were carried out in compliance with the current laws governing genetic experimentation in the Czech Republic.

References

Alkhimova OG, Mazurok NA, Potapova TA, Zakian SM, Heslop-Harrison JS, Vershinin AV (2004) Diverse patterns of the tandem repeats organization in rye chromosomes. Chromosoma 113:42–52

Baranczewski P, Nehls P, Rieger R, Pich U, Rajewsky MF, Schubert I (1997) Formation and repair of O^6-methylguanine in recombination hot spots of plant chromosomes. Environ Mol Mutagen 29:394–399

Charlesworth B, Sniegowski P, Stephan W (1994) The evolutionary dynamics of repetitive DNA in eukaryotes. Nature 371: 215–220

Crooks GE, Hon G, Chandonia JM, Brenner SE (2004) WebLogo: a sequence logo generator. Genome Res 14:1188–1190

Dechyeva D, Gindullis F, Schmidt T (2003) Divergence of satellite DNA and interspersion of dispersed repeats in the genome of the wild beet *Beta procumbens*. Chromosome Res 11:3–21

Dellaporta SL, Wood J, Hicks JB (1983) A plant DNA minipreparation: version II. Plant Mol Biol Rep 1:19–21

Dover G (1982) Molecular drive: a cohesive mode of species evolution. Nature 299:111–116

Dover G (2002) Molecular drive. Trends Genet 18:587–589

Elder JFE, Turner BJ (1995) Concerted evolution of repetitive DNA sequences in eukaryotes. Q Rev Biol 70:297–320

Gazdová B, Široký J, Fajkus J, Brzobohatý B, Kenton A, Parokonny A, Heslop-Harrison JS, Palme K, Bezděk M (1995) Characterization of a new family of tobacco highly repetitive DNA, GRS, specific for the *Nicotiana tomentosiformis* genomic component. Chromosome Res 3:245–254

Grebenstein B, Grebenstein O, Sauer W, Hemleben V (1996) Distribution and complex organization of satellite DNA sequences in Aveneae species. Genome 39:1045–1050

Hall SE, Luo S, Hall AE, Preuss D (2005) Differential rates of local and global homogenization in centromere satellites from Arabidopsis relatives. Genetics 170:1913–1927

Herzel H, Weiss O, Trifonov EN (1999) 10–11 bp periodicities in complete genomes reflect protein structure and DNA folding. Bioinformatics 15:187–193

Heslop-Harrison JS, Murata M, Ogura Y, Schwarzacher T, Motoyoshi F (1999) Polymorphisms and genomic organization of repetitive DNA from centromeric regions of Arabidopsis chromosomes. Plant Cell 11:31–42

King K, Jobst J, Hemleben V (1995) Differential homogenization and amplification of two satellite DNAs in the genus *Cucurbita* (Cucurbitaceae). J Mol Evol 41:996–1005

Kubis S, Schmidt T, Heslop-Harrison JS (1998) Repetitive DNA elements as a major component of plant genomes. Ann Bot 82:45–55

Langdon T, Seago C, Jones RN, Ougham H, Thomas H, Forster JW, Jenkins G (2000) De novo evolution of satellite DNA on rye B chromosome. Genetics 154:869–884

Lim KY, Skalická K, Koukalová B, Volkov RA, Matyášek R, Hemleben V, Leitch AR, Kovařík A (2004) Dynamic changes in the distribution of a satellite homologous to intergenic 26-18S rDNA spacer in the evolution of *Nicotiana*. Genetics 166:1935–1946

Linardopoulou EV, Williams EM, Fan Y, Friedman C, Young JM, Trask BJ (2005) Human subtelomeres are hot spots of interchromosomal recombination and segmental duplication. Nature 437:94–100

Macas J, Požárková D, Navrátilová A, Nouzová M, Neumann P (2000) Two new families of tandem repeats isolated from genus *Vicia* using genomic self-priming PCR. Mol Gen Genet 263:741–751

Macas J, Mészáros T, Nouzová M (2002) PlantSat: a specialized database for plant satellite repeats. Bioinformatics 18:28–35

Macas J, Navrátilová A, Mészáros T (2003) Sequence subfamilies of satellite repeats related to rDNA intergenic spacer are differentially amplified on *Vicia sativa* chromosomes. Chromosoma 112:152–158

Matyášek R, Gazdová B, Fajkus J, Bezděk M (1997) NTRS, a new family of highly repetitive DNA specific for the T1 chromosome of tobacco. Chromosoma 106:369–379

Meyne J, Goodwin EH (1995) Direction of DNA sequences within chromatids determined using strand-specific FISH. Chromosome Res 3:375–378

Morgante M, Jurman I, Shi L, Zhu T, Keim P, Rafalski JA (1997) The STR120 satellite DNA of soybean: organization, evolution and chromosomal specificity. Chromosome Res 5:363–373

Navrátilová A, Neumann P, Macas J (2003) Karyotype analysis of four *Vicia* species using in situ hybridization with repetitive sequences. Ann Bot 91:921–926

Navrátilová A, Neumann P, Macas J (2005) Long-range organization of plant satellite repeats investigated using strand-specific FISH. Cytogenet Genome Res 109:58–62

Nouzová M, Kubaláková M, Doleželová M, Koblížková A, Neumann P, Doleel J, Macas J (1999) Cloning and characterization of new repetitive sequences in field bean (*Vicia faba* L.). Ann Bot 83:535–541

Pedrosa A, Jantsch MF, Moscone EA, Ambros PF, Schweizer D (2001) Characterization of pericentromeric and sticky intercalary heterochromatin in *Ornithogalum longibracteatum* (Hyacinthaceae). Chromosoma 110:203–213

Sanger F, Nicklen D, Coulson AR (1977) DNA sequencing with chain terminating inhibitors. Proc Natl Acad Sci U S A 74:5463–5467

Schindelhauer D, Schwarz T (2002) Evidence for a fast, intrachromosomal conversion mechanism from mapping of nucleotide variants within a homogeneous α-satellite DNA array. Genome Res 12:1815–1826

Schmidt T, Heslop-Harrison JS (1998) Genomes, genes and junk: the large-scale organization of plant chromosomes. Trends Plant Sci 3:195–199

Schneider TD, Stephens RM (1990) Sequence Logos: a new way to display consensus sequences. Nucleic Acids Res 18: 6097–6100

Schubert I, Rieger R, Fuchs J, Pich U (1994) Sequence organization and the mechanism of interstitial deletion clustering in a plant genome (*Vicia faba*). Mutation Res 325:1–5

Shibata F, Hizume M, Kuroki Y (2000) Molecular cytogenetic analysis of supernumerary heterochromatic segments in *Rumex acetosa*. Genome 43:391–397

Skalická K, Lim KY, Matyášek R, Matzke M, Leitch AR, Kovařík A (2005) Preferential elimination of repeated DNA sequences from the parental, *Nicotiana tomentosiformis* genome donor of a synthetic, allotetraploid tobacco. New Phytol 166:291–303

Staden R (1996) The Staden sequence analysis package. Mol Biotechnol 5:233–241

Stephan W (1989) Tandem-repetitive noncoding DNA: forms and forces. Mol Biol Evol 6:198–212

Stephan W, Cho S (1994) Possible role of natural selection in the formation of tandem-repetitive noncoding DNA. Genetics 136:333–341

Tek A, Song J, Macas J, Jiang J (2005) Sobo, a recently amplified satellite repeat of potato, and its implications for the origin of tandemly repeated sequences. Genetics 170:1231–1238

Thompson JD, Higgins DG, Gibson TJ (1994) CLUSTAL W: improving the sensitivity of progressive multiple sequence alignment through sequence weighting, position-specific gap penalties and weight matrix choice. Nucleic Acids Res 22:4673–4680

Ugarkovic D, Plohl M (2002) Variation in satellite DNA profiles—causes and effects. EMBO J 21:5955–5959

Vershinin AV, Heslop-Harrison JS (1998) Comparative analysis of the nucleosomal structure of rye, wheat and their relatives. Plant Mol Biol 36:149–161